

Supplementary Material for Remark 4: Detailed Derivation of the $\mathcal{H}_\beta$ -Condition and ISS Verification for Reset Systems

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1 Introduction and Problem Formulation

In the stability analysis of hybrid reset control systems, the fundamental challenge is to find a common quadratic Lyapunov function $V(\chi) = \chi^\top P \chi$ (with $P = P^\top > 0$) that simultaneously satisfies two inherently conflicting Linear Matrix Inequalities (LMIs):

1. **Continuous Flow LMI:** The energy must strictly decay during continuous-time evolution, characterized by

$$A_\chi^\top P + P A_\chi \leq -\lambda_0 P, \quad \text{for some } \lambda_0 > 0. \quad (1)$$

2. **Discrete Jump LMI:** The energy must not increase across discrete reset events, characterized by

$$A_{\chi,\rho}^\top P A_{\chi,\rho} - P \leq 0, \quad (2)$$

where $A_{\chi,\rho}$ is the reset jump matrix mapping the pre-jump state to the post-jump state ($\chi^+ = A_{\chi,\rho} \chi$).

Solving these LMIs directly is computationally restrictive. The $\mathcal{H}_\beta$ -condition [1] circumvents this by translating the structural matrix-search problem into a straightforward frequency-domain Strict Positive Realness (SPR) check.

2 The $\mathcal{H}_\beta$ -Condition and the KYP Lemma

2.1 Artificial System Construction

In the standard application of the $\mathcal{H}_\beta$ -condition [1], the closed-loop state vector χ is decomposed into three distinct blocks: plant states (dimension n), resetting controller states (dimension $n_\rho = 1$), and non-resetting controller states (dimension $n_{\bar{\rho}}$).

An artificial strictly proper transfer function $H_\beta(s) = C_{h_\beta}(sI - A_\chi)^{-1}B_{h_\beta}$ is formulated using the following synthesized matrices:

$$B_{h_\beta} = \begin{bmatrix} 0_{n \times n_\rho} \\ I_{n_\rho} \\ 0_{n_{\bar{\rho}} \times n_\rho} \end{bmatrix}, \quad C_{h_\beta} = [\beta C \quad P_\rho \quad 0_{n_\rho \times n_{\bar{\rho}}}], \quad (3)$$

where $\beta \in \mathbb{R}^{n_\rho \times p}$ is a free matrix and $P_\rho = P_\rho^\top > 0$ is a chosen positive-definite matrix representing the target Lyapunov block for the reset states.

Definition 1 (The $\mathcal{H}_\beta$ -Condition [1]). *The $\mathcal{H}_\beta$ -condition is satisfied if there exists a matrix β and a matrix $P_\rho > 0$ such that the transfer function $H_\beta(s)$ is Strictly Positive Real (SPR), meaning $\text{Re}\{H_\beta(j\omega)\} > 0$ for all $\omega \in \mathbb{R} \cup \{\infty\}$.*

2.2 The Kalman–Yakubovich–Popov (KYP) Lemma

The theoretical foundation of the $\mathcal{H}_\beta$ -condition relies on the strict KYP lemma (see, e.g., [2, Lemma 6.3]). The KYP lemma states that $H_\beta(s)$ is SPR **if and only if** there exists a symmetric positive-definite matrix $P = P^\top > 0$ that satisfies the following algebraic constraints:

$$A_\chi^\top P + PA_\chi < 0, \quad (4)$$

$$PB_{h_\beta} = C_{h_\beta}^\top. \quad (5)$$

By verifying that $H_\beta(s)$ is SPR in the frequency domain, the KYP lemma mathematically guarantees the existence of a matrix P that inherently satisfies both (4) and (5). The subsequent sections illustrate how these two outputs uniquely satisfy the continuous flow and discrete jump LMIs of the reset system.

3 Derivation of the Strict Continuous Flow Bound (λ_0)

The KYP lemma guarantees the strict inequality (4). However, robust control against matched disturbances requires the formulation in (1), $A_\chi^\top P + PA_\chi \leq -\lambda_0 P$. This strict scalar bound λ_0 can be mathematically derived step-by-step as follows.

Step 1: Define the positive-definite matrix Q .

From the KYP lemma, it is established that $A_\chi^\top P + PA_\chi < 0$. Let Q be defined as:

$$Q = -(A_\chi^\top P + PA_\chi). \quad (6)$$

Because of the strict inequality, Q is symmetric and strictly positive-definite ($Q > 0$).

Step 2: Formulate the objective inequality.

The objective is to find a scalar $\lambda_0 > 0$ such that $A_\chi^\top P + PA_\chi \leq -\lambda_0 P$. Substituting Q , this is precisely equivalent to proving:

$$Q \geq \lambda_0 P. \quad (7)$$

Step 3: Apply a congruence transformation.

Because $P = P^\top > 0$, its unique principal matrix square root $P^{1/2}$ and its inverse $P^{-1/2}$ exist, and both are symmetric and positive-definite. In linear algebra, multiplying an inequality by $M^\top(\cdot)M$ preserves positive definiteness. Multiplying both sides of the target inequality by $M = (P^{-1/2})^\top = P^{-1/2}$ yields:

$$P^{-1/2}QP^{-1/2} \geq \lambda_0(P^{-1/2}PP^{-1/2}). \quad (8)$$

Step 4: Simplify the right-hand side.

Noting that $P = P^{1/2}P^{1/2}$, the right-hand side reduces identically to the identity matrix I :

$$P^{-1/2}PP^{-1/2} = P^{-1/2}(P^{1/2}P^{1/2})P^{-1/2} = I. \quad (9)$$

Thus, the inequality simplifies to:

$$P^{-1/2}QP^{-1/2} \geq \lambda_0 I. \quad (10)$$

Step 5: Utilize the minimum eigenvalue property.

By the Rayleigh quotient theorem, for any real symmetric matrix M , it holds that $M \geq \lambda_{\min}(M)I$. Let $M = P^{-1/2}QP^{-1/2}$. Because $Q > 0$, the transformed matrix M is also strictly positive-definite, meaning all of its eigenvalues are strictly positive. If λ_0 is chosen as:

$$\lambda_0 = \lambda_{\min}(M) = \lambda_{\min}(P^{-1/2}QP^{-1/2}) > 0, \quad (11)$$

then the inequality (10) is mathematically guaranteed to hold.

Step 6: Reconstruct the flow LMI.

By multiplying $M \geq \lambda_0 I$ on the left and right by $P^{1/2}$ (reversing the transformation), it follows that $Q \geq \lambda_0 P$. Substituting the definition of Q back into the inequality yields:

$$A_\chi^\top P + P A_\chi \leq -\lambda_0 P. \quad (12)$$

This confirms that the SPR property inherently satisfies the strict continuous flow LMI required for robust Input-to-State Stability (ISS).

4 Structural Forcing and the Discrete Jump LMI

While the flow LMI ensures stability during continuous time, the jump LMI requires P to have a specific block-diagonal structure so that energy does not increase during discrete resets. This structure is enforced by the second KYP equality, $P B_{h_\beta} = C_{h_\beta}^\top$.

By partitioning the unknown symmetric matrix P into a 3×3 block structure corresponding to the plant states (1), resetting controller states (2), and non-resetting controller states (3):

$$P = \begin{bmatrix} P_{11} & P_{12} & P_{13} \\ P_{12}^\top & P_{22} & P_{23} \\ P_{13}^\top & P_{23}^\top & P_{33} \end{bmatrix}. \quad (13)$$

Substituting P , along with the artificial matrices B_{h_β} and C_{h_β} from (3), into the KYP equality yields:

$$\begin{bmatrix} P_{11} & P_{12} & P_{13} \\ P_{12}^\top & P_{22} & P_{23} \\ P_{13}^\top & P_{23}^\top & P_{33} \end{bmatrix} \begin{bmatrix} 0 \\ I_{n_\rho} \\ 0 \end{bmatrix} = \begin{bmatrix} (\beta C)^\top \\ P_\rho^\top \\ 0 \end{bmatrix}. \quad (14)$$

Carrying out the matrix multiplication enforces three rigid constraints on P :

$$P_{12} = C^\top \beta^\top, \quad (15)$$

$$P_{22} = P_\rho, \quad (16)$$

$$P_{23}^\top = 0 \implies P_{23} = 0. \quad (17)$$

By setting the free parameter $\beta = 0$, equation (15) imposes $P_{12} = 0$. Consequently, the KYP lemma guarantees that the Lyapunov matrix P is perfectly block-diagonal with respect to the resetting states:

$$P = \begin{bmatrix} P_{11} & 0 & P_{13} \\ 0 & P_\rho & 0 \\ P_{13}^\top & 0 & P_{33} \end{bmatrix}. \quad (18)$$

5 Extension to Partial Resets: $\gamma \in (-1, 1)$

The standard reset mechanism instantaneously sets the controller state to zero ($\gamma = 0$). However, this can induce aggressive transient responses. Using a partial reset ratio $\gamma \in (-1, 1)$ allows for tunable damping.

It can now be shown that the identical Lyapunov matrix P derived for $\gamma = 0$ via the $\mathcal{H}_\beta$ -condition perfectly satisfies both LMIs across the entire continuous parameter range $\gamma \in (-1, 1)$.

5.1 Preservation of the Continuous Flow LMI

The continuous flow matrix A_χ represents the linear dynamics of the plant and the base controller. It governs the system strictly *between* reset events. Therefore, A_χ is mathematically independent of the discrete reset ratio γ . Because A_χ is unchanged, the flow condition derived previously remains identically satisfied:

$$A_\chi^\top P + PA_\chi \leq -\lambda_0 P, \quad \forall \gamma \in (-1, 1). \quad (19)$$

5.2 Preservation of the Discrete Jump LMI

The jump matrix for a partial reset is given by updating only the resetting state by a factor of γ , leaving the plant and non-resetting controller states untouched:

$$A_{\chi,\rho} = \begin{bmatrix} I_n & 0 & 0 \\ 0 & \gamma I_{n_\rho} & 0 \\ 0 & 0 & I_{n_{\bar{\rho}}} \end{bmatrix}. \quad (20)$$

Evaluating the discrete jump energy condition $A_{\chi,\rho}^\top P A_{\chi,\rho} - P$ utilizing the guaranteed structure of P from (18) gives:

$$A_{\chi,\rho}^\top P A_{\chi,\rho} - P = \begin{bmatrix} I & 0 & 0 \\ 0 & \gamma I & 0 \\ 0 & 0 & I \end{bmatrix} \begin{bmatrix} P_{11} & 0 & P_{13} \\ 0 & P_\rho & 0 \\ P_{13}^\top & 0 & P_{33} \end{bmatrix} \begin{bmatrix} I & 0 & 0 \\ 0 & \gamma I & 0 \\ 0 & 0 & I \end{bmatrix} - P. \quad (21)$$

Because the resetting block P_ρ is perfectly decoupled (the cross-terms are zero), the matrix multiplication does not mix the resetting and non-resetting blocks. The result evaluates cleanly to:

$$A_{\chi,\rho}^\top P A_{\chi,\rho} - P = \begin{bmatrix} P_{11} & 0 & P_{13} \\ 0 & \gamma^2 P_\rho & 0 \\ P_{13}^\top & 0 & P_{33} \end{bmatrix} - \begin{bmatrix} P_{11} & 0 & P_{13} \\ 0 & P_\rho & 0 \\ P_{13}^\top & 0 & P_{33} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & (\gamma^2 - 1)P_\rho & 0 \\ 0 & 0 & 0 \end{bmatrix}. \quad (22)$$

Because $P_\rho > 0$, the sign of the resulting matrix depends entirely on the scalar term $(\gamma^2 - 1)$. For any partial reset ratio strictly within the bounds $-1 < \gamma < 1$, it follows that:

$$\gamma^2 < 1 \implies (\gamma^2 - 1) < 0. \quad (23)$$

Thus, the block $(\gamma^2 - 1)P_\rho$ is negative semi-definite (and strictly negative definite for the reset channel if $\gamma \neq \pm 1$). The entire jump condition satisfies:

$$A_{\chi,\rho}^\top P A_{\chi,\rho} - P \leq 0, \quad \forall \gamma \in (-1, 1). \quad (24)$$

Conclusion: By formulating the problem via the $\mathcal{H}_\beta$ -condition and setting $\beta = 0$, the resulting structural constraints on the Lyapunov matrix P inherently decouple the resetting states from the rest of the system. This analytical decoupling mathematically guarantees that the robust hybrid stability established for standard resets ($\gamma = 0$) logically extends to all partial resets $\gamma \in (-1, 1)$ without requiring any modification to the underlying Lyapunov function.

References

- [1] O. Beker, C.V. Hollot, Y. Chait, and H. Han. Fundamental properties of reset control systems. *Automatica*, 40(6):905–915, 2004.
- [2] H. K. Khalil. *Nonlinear Systems*. Prentice Hall, Upper Saddle River, NJ, 3 edition, 2002.